Himadri Mandal

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Education

École Normale Supérieure, Paris **Septermber 2026**
D'ENS in Mathematics

Indian Statistical Institute, Kolkata **August 2023 — May 2026**
Statistics Undergraduate

Programming experience

Python: PyTorch, Numpy, Pandas, Bash, R, Next.js + TailwindCSS, LaTeX, Linux (Arch Linux on i3)

Work experience

Researcher: Truth Audit Labs, Delhi *June 2026 — July 2026*

Visiting Student Researcher: SCDLDS, Ashoka University *May 2025 — January 2026*

Research

Generalizing Hopf-Lax in Minmax Games: Sushant Vijayan *Ongoing (**TAL**)*
Using nonexpansive operators in hemi-metric spaces to generalize Hopf-Lax for minmax games with a subadditive initial function.

Fourier Entropy Influence Conjecture: Subhash Khot, **Courant** *Blog — Ongoing*

Hypercontractivity of Mixing Measures with Uncountable Spin States: Ayan Ghosh *Paper*

Does Order Matter: Robust Generalization: Mihir, Aritra et. al. ***Submitted to NeurIPS 2026***

Active Simple Hypothesis Testing: Sushant Vijayan, **TIFR** ***IEEE ISIT 2026***

Limits of Wtd Emp Samplers: Sarvesh R. Iyer, Sandeep Juneja, et. al. ***Poster, AISTATS 2026***

Guarantees for Diffusion on Tilted Samples: Sandeep Juneja, et. al. *arXiv*

Polynomial Sparsity Bounds: A. R. Chowdhury, V. Bhargava (**IIT Bombay**) *May 2025*
Research project on improving the $d^2 \log(n)$ bound in the Sparsity Bounds conjecture.

Circuit Phenomenology Using Sparse Autoencoders: David Udell, **SPAR** ***Preprint***

Selected Courses/Workshops

IEEE International Symposium of Information Theory, 2026: Guangzhou, China *July 2026*

exSPLORe2026 Workshop: **SCDLDS**, Ashoka University *January 2026*

DS: Probability and Optimization Methods: ICTS, Bangalore *August 2025*

exSPLORe2025 Workshop: **SCDLDS**, Ashoka University *January 2025*

CaMLAB: Cambridge AI Safety Hub *April 2024*

Deep Learning: ISI Kolkata *January 2024 — March 2024*

Measure Theory: Maths Club, ISI Kolkata *December 2023 — February 2024*

Projects

Theoretical

Independence Is Almost Dependence: *Blog.*

Axiom of Choice, the Zorn's Lemma and their equivalence: *Blog.*

Cold Reflections:
Link. My blog on Mathematics, Statistics, Philosophy and everything else that interests me. I add some of my work there.

Empirical/Programming

Ponderings on OthelloGPT: *Blog.*

ORIGAMI: *Repo.*

Selected fellowships and awards

ENS Paris Normalien: awarded April 2026, 1000€ per month (out of 4 selected internationals)

ISI K. Outstanding Performance: awarded January 2024, received ₹1500

ISI Kolkata B.Stat. Entrance: awarded August 2023, ranked 11th (out of 10,000 students)

Indian Olympiad Qualifier for Mathematics, KV: awarded February 2021, rank 4 in my region (out of >100,000 students all over India)

Atlas Fellowship India Finalist: awarded September 2022, received 1000$, top 200 (out of 4000 students) in a rationality fellowship

IISc Enumeration Finalist: awarded October 2022

CMI Tessellate Finalist: awarded October 2021

More experience

Volunteer and outreach

Owner: awas *October 2020 — October 2022*

Served as the **organizer and mentor** for daily math problem solving sessions, philosopical debates, and programming discussions for over two years on **Discord**. Mentored smart math enthusiasts, from all over India, learn hard math, and guided a few of them to get into the Indian training camp of IMO.

Owner: LML: Learning Machine Learning, ISI Kolkata *January 2024 — March 2025*

Organized extensive discussions on and around Machine Learning (Theory & Applied), Artificial Intelligence, and Statistics papers for more than a year.

Core Member: Theory CS Club, ISI Kolkata *March 2025 — Ongoing*

Engaged in extensive discussions on research papers on and around Theoretical CS focussing on Complexity Theory.